ONE-PAGE BRIEF

Post-quantum cryptography review: frost @ 0966bd1529aa

Client: Zcash Foundation — public repository (passive review) · Source: github.com/ZcashFoundation/frost, commit `0966bd1529aa` · Reviewed: 2026-11-04 18:47 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY
Level 0 of 4
no post-quantum cryptography

FINDINGS
0 Critical **0** High **4** Medium
3 Low **2** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)
Key establishment: 10 + 5 − 8 = 7 → **too late**
Signatures: 10 + 5 − 8 = 7 → **too late**

Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

WHAT MATTERS MOST

Medium QSI-2026-11933	**No hybrid post-quantum key exchange configured (book/src/zcash/server.md)** What to do: With OpenSSL ≥ 3.5 (or BoringSSL/rustls with ML-KEM), add X25519MLKEM768 ahead of X25519 in the groups list after interoperability testing.
Medium QSI-2026-11934	**Quantum-vulnerable FROST: threshold signing and MPC protocols (27 file(s))** preliminary What to do: Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the signature scheme; record which keys…
Medium QSI-2026-11935	**Quantum-vulnerable ECDSA: threshold signing and MPC protocols (2 file(s))** preliminary What to do: Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the signature scheme; record which keys…
Medium QSI-2026-11936	**Quantum-vulnerable ECDSA: role not determined (1 file(s))** preliminary What to do: Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form…

3 more in the full report.

WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

Role	Algorithms	What to do
Medium threshold signing and MPC protocols	FROST, ECDSA	Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the…
Medium role not determined	ECDSA	Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205)…

KNOWN VULNERABLE COMPONENTS

Advisories for the versions in use: 4; unlikely to apply to this build: 1 (a Cargo feature that is off, affected functions not named in the code, or a platform that is not built); to review: 3.

Static review of source code and manifests; files analysed: 61. No requests were sent to the owner's systems.

Full report, content SHA-256: `7afc256e62bd9d240cd3cbca50c07193224c5f4d93f1c709040c255693a230ad`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.